

A correlational study between human attention and high-level image perception

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Journal of Vision August 2017, Vol.17, 705. doi:<https://doi.org/10.1167/17.10.705>

Abstract

People have a remarkable ability to attend selectively to one or a few sensory inputs while ignoring the other ones (J. Moran, & R. Desimone, 1985). Although many studies have been made on selective attention at object level (T. Brosch, G. Pourtois, & D. Sander, 2010), less is known about the human attention and visual perception on complex natural scenes. We studied the correlation between human attention and visual perception of images of natural scenes. Sixteen subjects (mean age = 27) freely viewed 1249 emotional images and had their eye movements recorded. Another group of 358 participants viewed the same set of images and annotated a comprehensive list of 33 scene-level attributes, which includes 10 emotions (happiness, surprise, awe, excitement, amusement, contentment, sadness, anger, fear, and disgust) and 23 other attributes commonly studied in computer science community (e.g., aesthetics, image quality). Analyses indicated the following relationships among observer fixation patterns and image attributes: (1) Human has generally longer fixation duration and saccade duration on images with positive sentiments (e.g., aesthetics, awe), but shorter fixation on images with negative sentiments (e.g., sad, disgust). (2) Human has a general shorter saccade length and lower saccade velocity on images that are more centered and symmetric. (3) When an image is of high quality or

having focused object, it will usually lead to shorter saccade duration and higher saccade velocity of its observer. In summary, we found that image semantics, sentiment, and spatial layout correlate with human fixation patterns in a significant way. Our method is general and comprehensive in the sense that it focuses on complex natural scenes and studies on an intensive attribute lists.

Meeting abstract presented at VSS 2017

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A correlational study between human attention and high-level image perception (supplementary material)

This supplementary material provides additional experimental details and examples to complement the abstract.

1. Examples of emotional images

Our emotional image set is collected from two sources: (1) a subset (389) photos of the International Affective Picture System (IAPS) (P. Lang, M. Bradley, & B. Cuthbert, 2008); (2) a set of 860 photos collected by the authors. We grouped the images into six types based on how they evoked emotions in viewers (see Fig. 1 for examples).

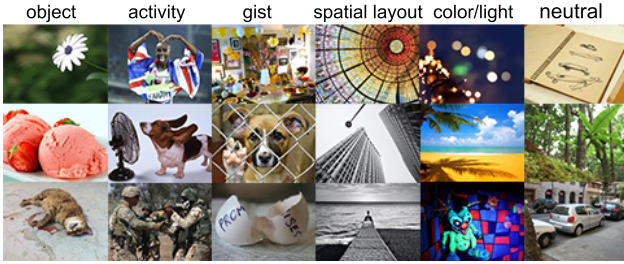


Figure 1: Example images illustrating the types based on how they evoked emotions in viewers.

2. Scene-level attributes annotation

We built an online scene-annotation platform to collect perceptions of 33 high-level image attributes. Each image was rated on 33 high-level image attributes with a 9-point scale. The detailed list of attributes is shown in Table 1.

3. Analysis and Results

We investigated the correlation between high-level attributes and human fixations in the time scale. We first dichotomized each of the attribute ratings (which ranges between 0 and 1) to binary labels by setting a threshold of .5 (e.g., an image with a rating higher than .5 on happy will be positive on this attribute, negative otherwise). We used 4 statistics commonly used in fixation analysis, namely fixation duration, saccade duration, saccade length, and saccade velocity. Fig. 2 shows examples of the results.

Type	Detailed attributes
Emotions	Happy; Surprise; Awe; Excitement; Amusement; Contentment; Sad; Anger; Fear; Disgust.
SAM (M. Bradley & P. Lang, 1994)	Valence; Arousal; Dominance.
Semantics	Familiar; Unusual; Dynamic; Informative; Natural object combination.
Aesthetics	Aesthetic; High quality; Colorful; Natural color combination; Sharp.
Spatial layout	Have objects of focus; Single focused; Close-up shot; Centered; Symmetrical.
Naturalness	Photorealistic.
People related	Attractive person; Posing; Eye contact; Positive expression.

Table 1: List of 33 scene-level attributes.

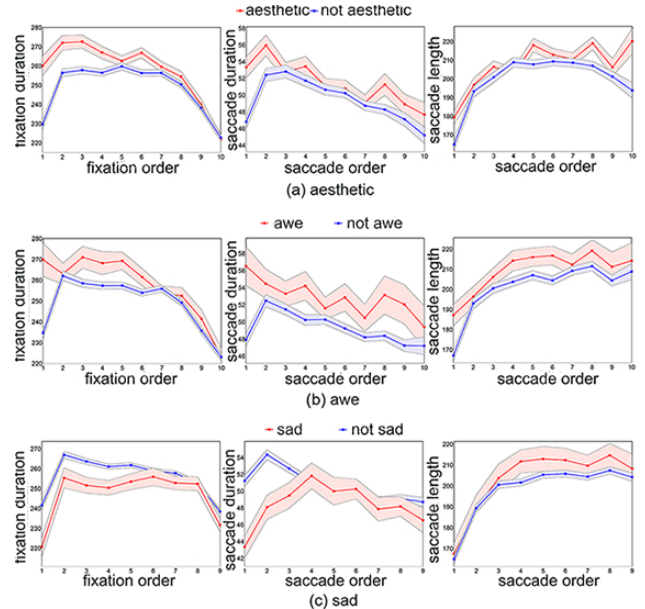


Figure 2: Fixation patterns correlate with attributes “aesthetic”, “awe”, and “sad” ($p < .001$).